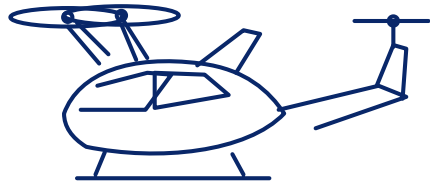


DESIGN MODEL

1

Design space

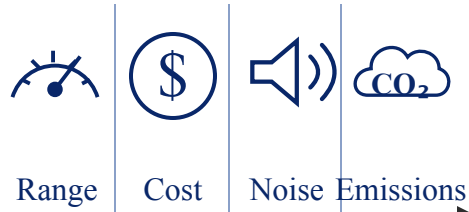


$$x \in X$$

2

MDO model

$$x \rightarrow y(x)$$



Normalize

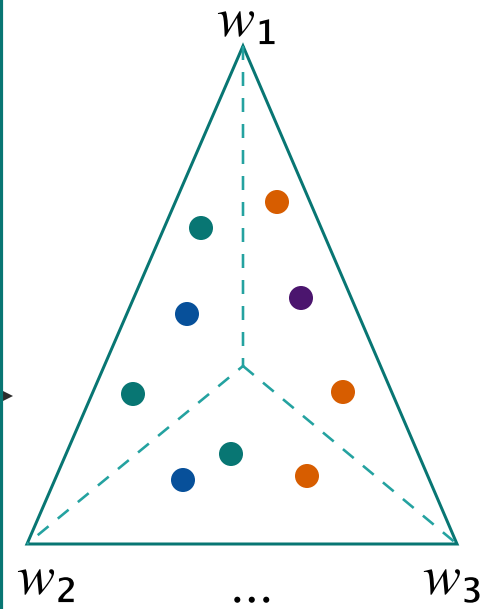
$$u(x) \in [0,1]^K$$

PREFERENCE UNCERTAINTY

3

Valuation futures

$$t_0 \longrightarrow T$$

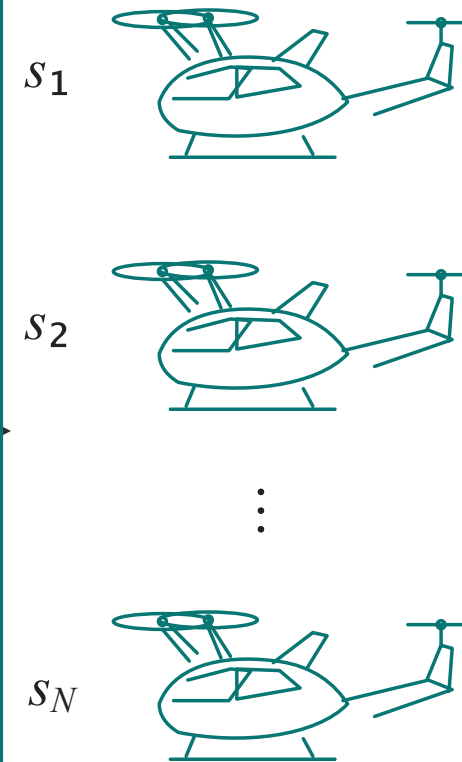


$$w \in W$$

$$\sum w_k = 1$$

4

Scenario optimization



$$s_i = \arg \max_x v(w_i, u(x))$$

ROBUST SELECTION

6

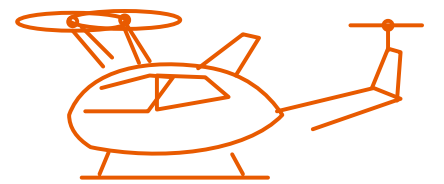
Robust selection



Consensus ranking

Sensitivity

Robust design commitment



s^*

5

Cross-scenario evaluation

$$V = [v_{ij}]$$

	w_1	w_2	...	w_N
s_1				
s_2				
$\vdots$				
s_N				

$$R = [r_{ij}]$$

	w_1	w_2	...	w_N
s_1				
s_2				
$\vdots$				
s_N				